

Galvanic Corrosion + Rust Removal

Lecture & Instructor Notes

Audience: instructors and teaching assistants

Purpose: Teaching script, deep-dive notes, and coaching guidance

Session hook: Corrosion, sacrificial protection, and reversing rust

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Lecture talking points
3. Instructor deep-dive notes
4. Preparation reminders

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 5 — Galvanic Corrosion and Electrolytic Rust Removal

Hook: Connect corrosion, sacrificial protection, and reversing rust with electricity.

Learning outcomes

- Build metal-pair galvanic cells and measure voltage
- Explain galvanic corrosion and sacrificial protection
- Interpret overnight electrolytic rust removal (started Session 4)
- Synthesize the week's experiments in a final review

Session plan (90 min)

Time	Activity	Reference
0–15 min	Hook: reveal derusted nails + corrosion jars	experiment.md — Part C
15–30 min	Galvanic cells, corrosion, sacrificial anodes	lecture.md
30–50 min	Metal-pair cells — measure voltage	experiment.md — Part A
50–60 min	Corrosion jar observations	experiment.md — Part B
60–75 min	Week synthesis: spontaneous vs driven cells	lecture.md
75–90 min	Final challenge questions	lecture.md

Key reactions

Galvanic: $\text{Zn(s)} \rightarrow \text{Zn}^{2+} + 2\text{e}^{-}$

Electrolytic derusting (cathode): rust reduced toward Fe; H_2 may evolve

Sacrificial anode (+): metal oxidizes

Multi-day setup

Demo	Start	Reveal
Corrosion jars A/B/C	Session 1–2	Session 5
Electrolytic rust removal	Session 4 (end)	Session 5 (opening)

Status

- [] Rust-removal cells started Session 4 end-of-class
- [] Metal samples prepared (Zn, Fe, Cu, Al)
- [] Final review questions prepared

- ☐ Washing soda or baking soda procured

Session 5 — Lecture: Corrosion, Galvanic Cells, and Electrolytic Rust Removal

Target duration: ~20 minutes (+ 10 min synthesis at end)

Instructors: activity series, galvanic couples, rust chemistry, derusting polarity — instructor.md.

Opening hook (0–15 min)

Conducted with experiment.md Part C — reveal overnight derusting cells.

Display:

- Treated vs. control rusty nails (from Session 4 setup)
- Corrosion jars A/B/C
- Shiny galvanized nail
- Copper wire
- Optional: photo of ship hull with sacrificial anode

"Last session we used electricity to split water. Overnight we used the same idea to undo rust. Why does iron rust faster when it touches copper — and how can we force the reaction the other way?"

Core concepts (15–30 min)

1. Galvanic cells — back to Session 1

When two different metals connect in an **electrolyte** (moisture, saltwater):

- More **active** metal → **anode** (oxidizes, corrodes)
- Less active metal → **cathode** (protected)
- Electrons flow through the metal contact → measurable **voltage**

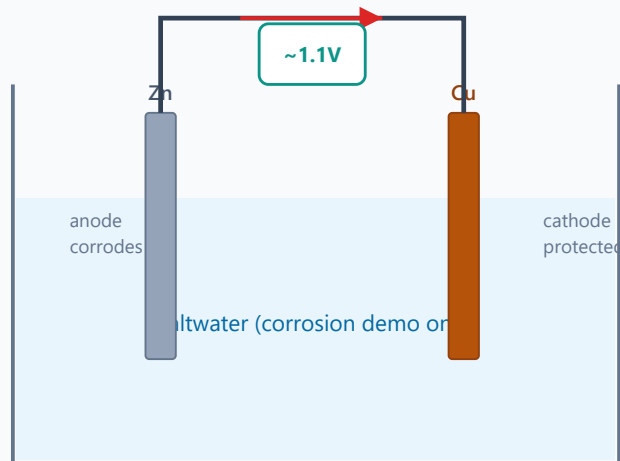
Activity series snippet (more reactive → oxidizes first):

Mg > Al > Zn > Fe > Cu (simplified for class)

2. Corrosion

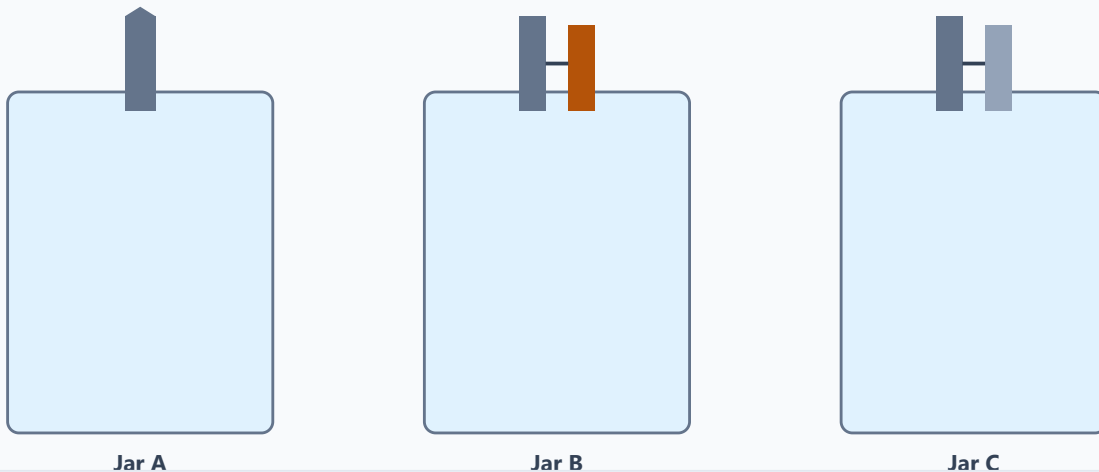
- Rust = iron oxidation in presence of water and oxygen
- **Galvanic corrosion:** iron + copper in saltwater → iron corrodes **faster**
- **Sacrificial protection:** zinc corrodes instead of iron (galvanizing, hull anodes)

Session 5 - Galvanic cell (Zn + Cu in saltwater)



$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ (oxidation at anode) | Spontaneous - produces electricity like Session 1
More active metal becomes anode; less active metal becomes cathode

Session 5 - Corrosion comparison jars



Galvanic coupling: when dissimilar metals touch in electrolyte, the more active metal corrodes preferentially

3. Sacrificial anodes

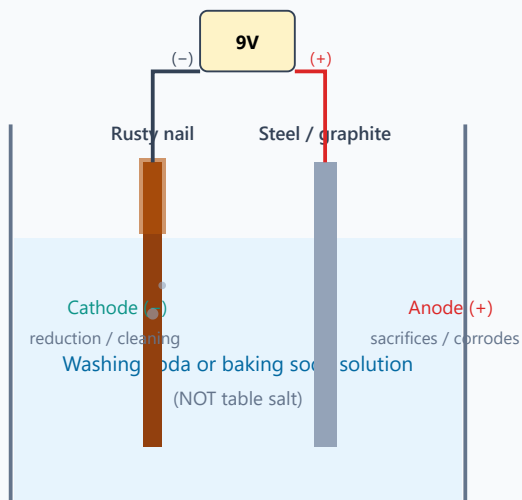
- Blocks of zinc/magnesium attached to steel structures
- Zinc is more active $\rightarrow$ becomes anode $\rightarrow$ corrodes preferentially
- Used on ships, pipelines, water heaters

4. Electrolytic rust removal — reversing corrosion

	Galvanic corrosion	Electrolytic derusting
Driven by	Spontaneous metal activity difference	External power supply

	Galvanic corrosion	Electrolytic derusting
Iron	Corrodes (anode in couple)	Cathode — rust reduced
Partner metal	Copper cathode protects itself	Sacrificial anode corrodes
Electrolyte	Often saltwater (demo)	Washing soda or baking soda — not NaCl

Electrolytic rust removal (start Session 4, reveal Session 5)



Start at end of Session 4; rinse and compare to control nail at start of Session 5
Same electrolysis idea as Session 4 - electricity drives a non-spontaneous reaction

Key idea: Same electrolysis principle as Session 4 — electricity forces a non-spontaneous reaction. At the cathode, rust is reduced toward clean iron; H_2 may evolve.

5. Fuel cells — brief concept (optional, no demo)

Session 4 stored energy in H_2 . A **fuel cell** could convert $H_2 + O_2$ back to electricity + water (reverse of electrolysis). Hydrogen is an **energy carrier** — round-trip efficiency is always $< 100\%$. We skip the PEM kit demo; the rust-removal cell shows the same "electricity drives chemistry" idea on a visible timescale.

Synthesis — Final challenge (80–90 min)

Ask the class (whole-group or worksheet):

1. Which experiment **made** electricity?
2. Which **consumed** electricity?
3. Which **moved metal atoms**?
4. Which **split molecules**?
5. Which explained **corrosion**?
6. Which **reversed** corrosion?
7. Where did **electrons** flow? Where did **ions** flow?

8. What was **oxidized**? What was **reduced**?
9. Which experiment is most relevant to **renewable energy**?

Answer key (instructor)

Question	Session(s)
Made electricity	1, 5 (galvanic)
Consumed electricity	2, 3, 4, 5 (rust removal)
Moved metal atoms	2, 3
Split molecules	4
Corrosion	5
Reversed corrosion	5 (electrolytic derusting)
Renewable energy link	4 (H ₂ storage concept)

Visual aids

- ☐ Activity series chart with arrows showing anode/cathode
- ☐ Diagram: ship with sacrificial anode
- ☐ Before/after photos of derusted nails
- ☐ Week-at-a-glance timeline (5 sessions)

Vocabulary (week review)

- ☐ Galvanic vs. electrolytic cell
- ☐ Sacrificial anode
- ☐ Energy carrier
- ☐ Oxidation / reduction (every session)

Closing message

"Electrochemistry is everywhere: phone batteries, car rust, chrome plating, hydrogen buses, and the lemon you measured on day one. Same ideas — electrons, ions, and the drive toward lower energy."

Session 5 — Instructor Notes: Corrosion, Galvanic Cells & Rust Removal Deep Dive

Audience: instructors and TAs only. Pair with lecture.md and experiment.md.

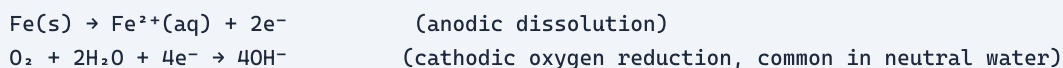
Why this session exists

This is the synthesis day. Students reconnect Session 1 (spontaneous galvanic cells) to real-world corrosion, see sacrificial protection, and open overnight electrolytic rust-removal cells started in Session 4. Your job: make **activity series**, **galvanic couples**, and **driven vs spontaneous** feel like one coherent story — not three demos.

Atomic foundations of corrosion

What rust is

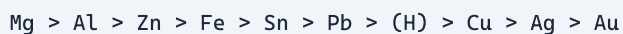
Iron metal oxidizes in the presence of water and oxygen. A simplified teaching sequence:



Fe^{2+} further oxidizes and precipitates as hydrated oxides / oxyhydroxides — the red-brown **rust** students recognize ($\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$ / FeOOH family). Exact mineralogy is complex; students need: *rust is oxidized iron, not dirt sitting on iron*.

Valence electrons and why iron corrodes

Iron ($Z = 26$) readily loses valence electrons to form $\text{Fe}^{2+}/\text{Fe}^{3+}$. In the activity series, Fe sits **above** Cu and **below** Zn:



Consequences you must own:

- Fe + Cu in saltwater → Fe becomes anode → **faster rust**
- Fe + Zn in saltwater → Zn becomes anode → **Fe protected** (galvanizing logic)
- Pure Fe still rusts alone (local anodes/cathodes on the same piece from impurities, stress, O_2 gradients)

Electrochemical series — instructor reference table

Approximate standard reduction potentials (acidic, vs SHE):

Half-reaction	E° (V)	Implication
$\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}$	-2.37	Very active; strong sacrificial candidate

Half-reaction	E° (V)	Implication
$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	-1.66	Active but often passive (oxide film)
$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$	-0.76	Classic sacrificial anode / galvanizing
$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$	-0.44	Rusts readily when coupled to Cu
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	0.00	Reference
$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	+0.34	Noble relative to Fe; accelerates Fe corrosion when coupled
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$	+0.80	Even more noble
$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$	+1.23	Key cathodic reaction in wet corrosion

Teaching rule of thumb: the metal with the **more negative** reduction potential tends to oxidize when two metals are coupled in an electrolyte.

Galvanic cells revisited (metal-pair lab)

What students measure

Two dissimilar metals in saltwater + multimeter $\approx$ a crude galvanic cell:

- More active metal $\rightarrow$ anode (corrodes)
- Less active $\rightarrow$ cathode (protected)
- Voltage polarity tells you which way electrons want to flow

Expected qualitative voltages (order of magnitude)

Pair	Typical open-circuit trend
Zn–Cu	Highest of common set (~ 1 V class)
Al–Cu	Can be high, but Al oxide may suppress
Fe–Cu	Medium
Zn–Fe	Smaller than Zn–Cu

Do not grade students on matching textbook E° exactly — saltwater concentrations, surface films, and area ratios dominate.

Saltwater role

NaCl solution:

- Provides ions for conduction
- Chloride promotes breakdown of passive films (especially relevant to real-world pitting)
- **Reminder:** saltwater is OK for corrosion demos; still **forbidden** for Session 4-style water electrolysis / derusting electrolytes

Corrosion jars A / B / C — narrative

Jar	Setup	What students should see	Story
A	Fe alone	Gradual rust	Local cells on one metal
B	Fe + Cu touching	Faster Fe rust	Galvanic couple; Fe anode, Cu cathode
C	Fe + Zn touching	Fe cleaner; Zn suffers	Sacrificial anode

Ship / water-heater analogy: zinc or magnesium blocks bolted to steel hulls / heaters are Jar C at engineering scale.

Why “touching” matters

Galvanic corrosion requires:

1. Electrical contact between metals (electron path)
2. Shared electrolyte (ion path)

If metals are insulated from each other, the couple does not form.

Electrolytic rust removal — deep dive

Polarity (repeat until automatic)

Piece	Battery terminal	Role
Rusty object	– cathode	Reduction / cleaning
Sacrificial steel or graphite	+ anode	Oxidizes / may sludge
Electrolyte	Washing soda (Na_2CO_3) or baking soda	Alkaline, no chloride

Why not NaCl

Chloride + anode can produce chlorine and aggressive pitting chemistry. Same rule as Session 4.

What “cleaning” means chemically

At the cathode, electrochemical reduction and vigorous local chemistry disrupt oxide layers; hydrogen evolution often helps lift rust. The object does not become “new steel” by magic — loosely bound oxides fall away; residual films may need brushing/rinsing. Results vary with rust thickness and overnight time.

Link to jars

Process	Iron’s role	Energy
Jar B galvanic corrosion	Anode (oxidizes)	Spontaneous
Electrolytic derusting	Cathode (reduced)	Driven by battery

This contrast is the intellectual climax of the week.

Reveal protocol (Session 5 opening)

1. Disconnect power.
2. Remove nails; rinse.
3. Compare to control and before photo.
4. Ask: which electrode was cathode? How is this like Session 4? How is it opposite of Jar B?

Sacrificial protection — mechanisms students can own

1. **Galvanizing:** zinc coating on steel. Zinc corrodes preferentially; even if scratched, Zn can still protect nearby steel electrochemically (within electrolyte contact).
2. **Impressed current / electrolytic protection:** external power forces structure to be cathodic (related idea to derusting polarity).
3. **Paint / barriers:** not electrochemical, but prevent electrolyte/O₂ access — mention as complementary.

Fuel cells — optional concept only

If asked:

Electrolysis: electricity + H₂O → H₂ + O₂
Fuel cell: H₂ + O₂ → electricity + H₂O

Same chemistry, opposite energy direction. No PEM kit required this week. Emphasize efficiency losses and “energy carrier” language from Session 4.

Week synthesis — how to run the final challenge

Have the answer key ready (lecture.md). Push students to use vocabulary:

- Galvanic vs electrolytic
- Anode / cathode
- Oxidation / reduction
- Electron path vs ion path

Strong closing: every session was about **where electrons go and why**.

Misconceptions

Misconception	Correction
“Rust is a coating that sits on iron.”	Rust is iron that has already oxidized .
“Copper makes iron rust by touching it chemically like paint.”	Copper makes iron the anode in a galvanic couple.
“Zinc coating just seals the steel.”	Sealing helps, but Zn also sacrifices electrochemically.

Misconception	Correction
"Derusting means the battery dissolves rust into nothing."	Electrochemistry + rinsing removes / reduces oxide; sludge remains in bath.
"Salt is always the best electrolyte."	Great for corrosion demos; dangerous for Cl_2 -producing electrolysis.

Board plan

1. Reveal derusted vs control nails (visual hook).
2. Activity series arrow; place Zn, Fe, Cu.
3. Draw jars A/B/C with anode labels.
4. Contrast galvanic rusting vs electrolytic cleaning table.
5. Final challenge worksheet → closing message.

Optional enrichment

- Crevice corrosion and differential aeration (why nails rust under washers).
- Stainless steel passivation (Cr_2O_3 film) — why "stainless" is not magic.
- Cathodic protection standards in marine engineering (names only).
- Pourbaix diagram for Fe (advanced TA interest).

Session 5 — Preparation

Before class (instructor)

- [] Confirm derusting cells were started Session 4 end-of-class (see Session 4 prep)
- [] Photograph nails before treatment if not done Thursday
- [] Prepare saltwater for galvanic demos (label "**corrosion only — not for gas electrolysis**")
- [] Clean metal strips; sand if oxidized
- [] Print final challenge worksheets

Session 4 handoff (critical)

Derusting cells must be running **before students leave Session 4**:

Item	Detail
Electrolyte	Washing soda preferred; baking soda OK — not NaCl
Cathode (–)	Rusty iron
Anode (+)	Sacrificial steel or graphite
Control	Rusty nail in plain water, no power
Power	9 V battery; secure clips; label each cell

Optional multi-day corrosion setup

Start Monday or Tuesday for best rust by Friday:

Jar	Setup
B	Iron nail wired/touching copper in saltwater
C	Iron nail wired/touching zinc in saltwater
A	Iron alone (control)

Store open jars where students can observe daily changes.

Day-of timeline

When	Task
0 min	Reveal derusted nails + corrosion jars (Part C)
25 min	Metal-pair galvanic cells (Part A)
45 min	Corrosion jar observations (Part B)

When	Task
70 min	Week comparison table (Part D)
80 min	Final challenge; collect worksheets

Rust removal troubleshooting

Issue	Fix
Little visible change overnight	Fresher rust; higher soda concentration; check connections
Anode heavily corroded	Expected — sacrificial behavior
Cloudy solution	Normal; rinse nail before comparing
Chlorine smell	Stop — salt was used; replace electrolyte

Open questions / notes

- Washing soda vs. baking soda used:
- Overnight run time (hours):
- Corrosion jars started on date: